

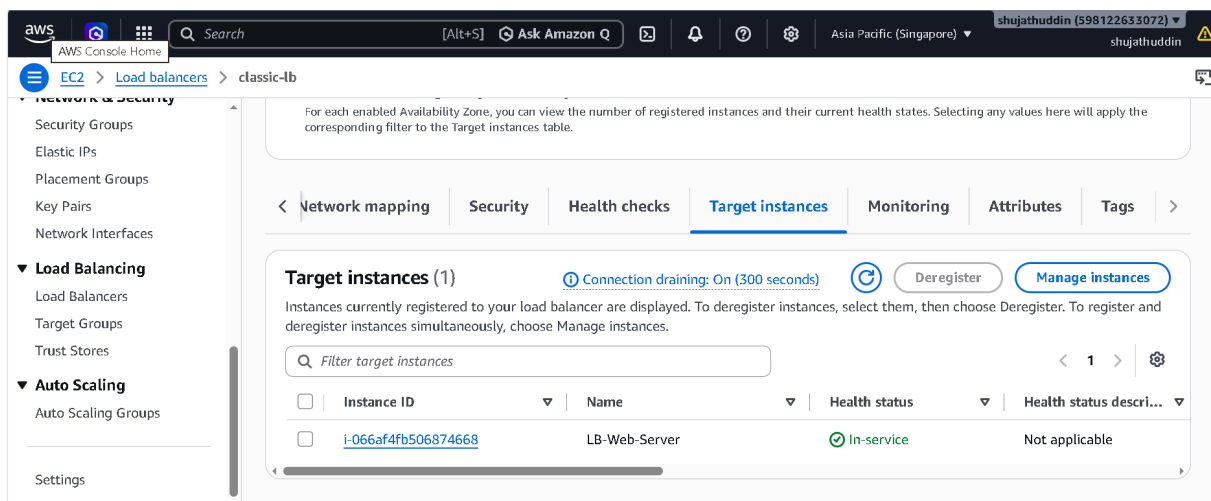
## 1. Configure Classic Load balancer.

- Created and configured a Classic Load Balancer.
- Registered EC2 instances as backend instances.
- Configured listeners and security groups.
- Verified that traffic was distributed across the registered instances



## 2. Configure Application Load balancer.

- Created an internet-facing Application Load Balancer.
- Configured public subnets and security groups.
- Created a target group and registered EC2 instances.
- Configured HTTP listener and health checks.
- Verified application access through the ALB DNS name



EC2 > Target groups > alb-targets

**alb-targets** Actions

**Details**

arn:aws:elasticloadbalancing:ap-southeast-1:598122633072:targetgroup/alb-targets/4519c15fe08fa850

|                                |                                                         |                                  |                                                     |
|--------------------------------|---------------------------------------------------------|----------------------------------|-----------------------------------------------------|
| <b>Target type</b><br>Instance | <b>Protocol : Port</b><br>HTTP: 80                      | <b>Protocol version</b><br>HTTP1 | <b>VPC</b><br><a href="#">vpc-0a5149f83eadefc6f</a> |
| <b>IP address type</b><br>IPv4 | <b>Load balancer</b><br><a href="#">None associated</a> |                                  |                                                     |

|                    |                             |                |             |              |               |
|--------------------|-----------------------------|----------------|-------------|--------------|---------------|
| 1<br>Total targets | 0<br>Healthy<br>0 Anomalous | 0<br>Unhealthy | 1<br>Unused | 0<br>Initial | 0<br>Draining |
|--------------------|-----------------------------|----------------|-------------|--------------|---------------|

**Distribution of targets by Availability Zone (AZ)**  
Select values in this table to see corresponding filters applied to the Registered targets table below.

EC2 > Load balancers > application-lb

**application-lb** Actions

**Details**

|                                          |                                      |                                                                                                                                                                                                                                                   |                                                              |
|------------------------------------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| <b>Load balancer type</b><br>Application | <b>Status</b><br>Active              | <b>VPC</b><br><a href="#">vpc-0a5149f83eadefc6f</a>                                                                                                                                                                                               | <b>Load balancer IP address type</b><br>IPv4                 |
| <b>Scheme</b><br>Internet-facing         | <b>Hosted zone</b><br>Z1LMS91P8CMLE5 | <b>Availability Zones</b><br><a href="#">subnet-0338e0fdaaaa476e0</a> ap-southeast-1c (apse1-az3)<br><a href="#">subnet-00e7cb2415d0f9c53</a> ap-southeast-1a (apse1-az2)<br><a href="#">subnet-0c7259c48b752f0e3</a> ap-southeast-1b (apse1-az1) | <b>Date created</b><br>September 15, 2026, 19:10 (UTC+05:30) |

← ↻ Not secure application-lb-809881758.ap-southeast-1.elb.amazonaws.com

Import favorites | Jira | Console Home | C... | GitHub | Untitled document...

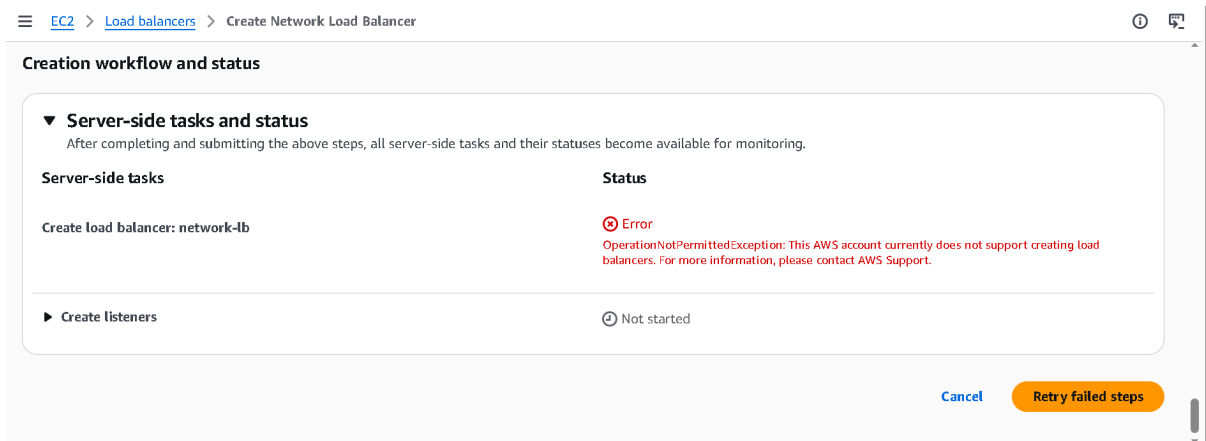
# Apache Web Server

## Load Balancer Practical

This page is running on the EC2 instance.

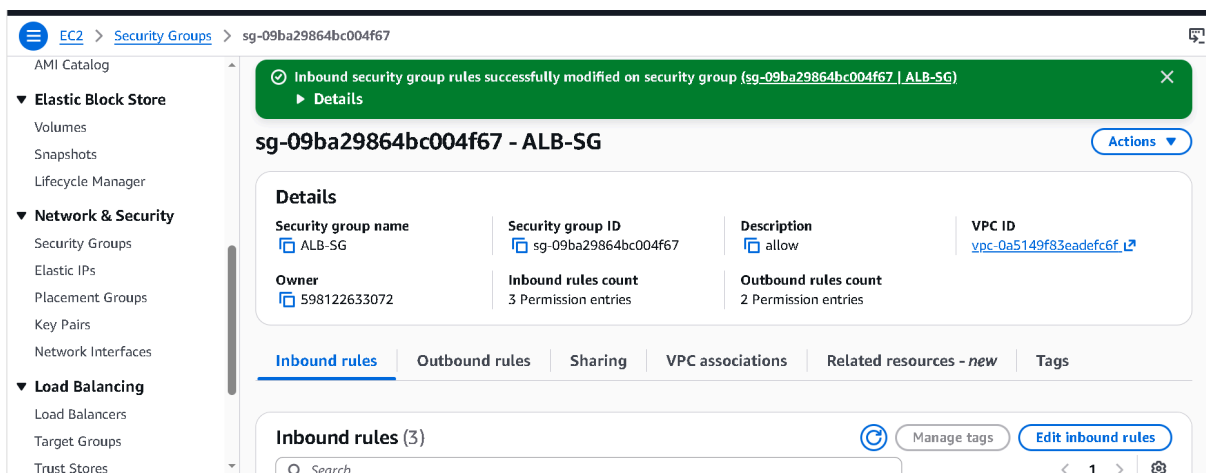
### 3. Configure Network Load balancer.

- Created a Network Load Balancer for high-performance TCP traffic.
- Configured appropriate subnets and listeners.
- Created a target group and registered backend EC2 instances.
- Verified connectivity through the NLB endpoint.



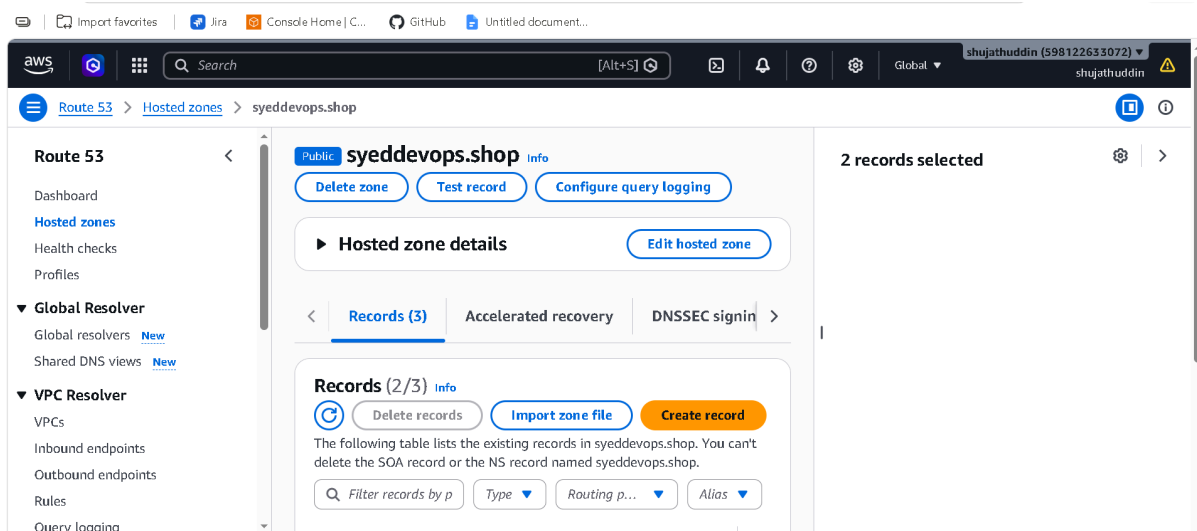
#### 4. Attach SSL for application load balancer.

- Requested/imported an SSL certificate using AWS Certificate Manager (ACM).
- Added an HTTPS listener on port **443** to the ALB.
- Associated the ACM certificate with the HTTPS listener.
- Configured HTTP-to-HTTPS redirection.
- Verified secure HTTPS access to the application.



#### 5. Map Application load balancer to R53.

- Created a hosted zone in Amazon Route 53.
- Created an **A record (Alias)** pointing to the Application Load Balancer.
- Mapped the domain name to the ALB.
- Verified that the application was accessible using the custom domain.



## 6. Push the application load balancer logs to S3.

- Created an S3 bucket for ALB access logs.
- Configured the required S3 bucket permissions.
- Enabled access logging on the Application Load Balancer.
- Specified the S3 bucket as the log destination.
- Verified that ALB log files were delivered to the S3 bucket.

### Logging

#### Access logs - legacy

S3 location: [application-lb-logs-2026-syed](#)

#### Connection logs - legacy

Off

#### Health check logs - legacy

Off

#### Logging options

Log delivery to CloudWatch, Kinesis Firehose, and S3 is configurable through an integration with Logging - by Amazon CloudWatch. Visit **Logging** in the [Integrations tab](#).

application-lb-logs-2026-syed [Info](#)

- <
- Objects
- Metadata
- Properties
- Permissions
- Metrics
- Management
- File systems
- A
- >

Objects (1)



Copy S3 URI

Copy URL

Download

Open ↗

Delete

Actions ▼

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

< 1 >



| <input type="checkbox"/> | Name ▲                                                                                                     | Type ▼ | Last modified ▼ | Size ▼ | Storage class ▼ |
|--------------------------|------------------------------------------------------------------------------------------------------------|--------|-----------------|--------|-----------------|
| <input type="checkbox"/> |  <a href="#">AWSLogs/</a> | Folder | -               | -      | -               |